Casey D. Foley, PhD

Laser & Vacuum Systems | Optical Metrology & Machine Vision | Spectroscopy | Process Development

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SUMMARY

Laser and vacuum systems expert and physical chemist with 12+ years across manufacturing, a national laboratory, and university research. Leads laser, vacuum, and metrology projects from idea to plant floor; results include 99% yield, 3× faster laser cycles, and a ~40% shorter vacuum process cycle.

CORE SKILLS

Lasers & Optics	Fiber, Nd:YAG/DPSS, CO2, excimer, Ti:sapphire, dye, IR OPO/OPA, VUV; drilling, ablation, heating; beam diagnostics, Zemax, Class 4 laser safety
Vacuum & Instruments	Production high vacuum, UHV design, cryogenics, plasma discharges, ion optics (SIMION), molecular beams, SolidWorks
Metrology & Vision	Deflectometry, confocal profilometry, line-scan vision, laser-based R-value measurement, GR&R, color measurement (CIELAB)
Spectroscopy	UV/Vis/IR, FTIR, REMPI, LIF, IR–UV double resonance, velocity-map imaging, IMS-MS
Process & Quality	Process development, FAT/SAT, commissioning, DOE, SPC, Cp/Cpk, PFMEA, 8D, control plans, SOPs, Lean/Six Sigma training
Software & Data	Python, JMP, LabVIEW, MATLAB; data acquisition, instrument control, machine learning
Communication	15+ conference presentations, 23 peer-reviewed papers, cross-functional liaison, mentoring and training

PROFESSIONAL EXPERIENCE

Staff Scientist — Apr 2026 – Present

Laser Systems Engineer — Aug 2024 – Mar 2026

LuxWall, Inc. — vacuum-insulated glass manufacturing

- **Technical Leadership:** Subject-matter expert for all laser and high vacuum processes and technical liaison between production, R&D, and scaling.
- **Laser Drilling:** Integrated a fiber laser through FAT and commissioning for a 3× faster cycle, and cut defects from ~1 in 10 parts to ≤1% with in-line confocal sensing and auto-focus.
- **Laser Ablation:** Doubled throughput of a second process at ~99% yield with in-line confocal depth metrology, closed-loop focus, vision-based part location, and SPC; set its depth spec from first principles.
- **Evacuation:** Conceived a physics-based process change, validated it through gated trials (DOE, FMEA, difference-in-differences), and aligned every level of the company; ~40% shorter cycle.
- **R-Value Metrology:** Developed a proprietary laser-based measurement from algorithm to plant floor, plus the protocol for production quality screening.
- **Flatness Metrology:** Led development of a phase-shifting deflectometry system (~10 µm) for in-line 100% inspection; wrote a Python pipeline for micron-level confocal profiling.
- **In-Line Inspection:** Wrote the spec, ran a competitive vendor selection, and led FAT/SAT of a ~50 µm/pixel line-scan system now measuring production lites and detecting defects; set its GR&R acceptance criteria and held the vendor to spec with tool-vs-manual correlation.
- **Process Capability:** Studied 14 QC characteristics (I-MR, Cp/Cpk), set control limits, revised spec limits, and quantified a ~70% variation cut from a sensor upgrade.
- **Root-Cause Analysis:** Eliminated drilling cracks using tens of thousands of production records (chi-square, random-forest importance), microscopy, and a new laser recipe.
- **Seal Quality:** Cut worst-rated seals from ~15% to ~3% with a new laser sealing recipe; wrote its reaction plan.
- **Capacity Planning:** Built a multi-line capacity simulator that engineering and management use to find constraints and test improvements.

- **Laser Safety & Standards:** Defined Class 4 controls (ANSI Z136.1/.9, FDA 21 CFR 1040, NFPA 115) for the production lasers; own control plans, preventive maintenance, SOPs, and work instructions.

Research Excellence Postdoctoral Fellow Aug 2022 – Jul 2024
University of Missouri — Department of Chemistry

- **Plasma Beam Source:** Designed and built a fast-atom/ion beam machine around a pulsed plasma-discharge source, from new and repurposed hardware, to study hypersonic-flow chemistry.
- **Plasma Diagnostics:** Characterized discharge plasmas (N^+, N_2^+, O^+, O_2^+, and Ar^+ beams) for two years with MCP/phosphor imaging, cameras, and PMTs.
- **VUV & Ultrafast:** Built a 118 nm VUV source (355 nm tripled in Xe/Ar); ran MeV ultrafast electron diffraction at SLAC with custom Python analysis.
- **Mentoring:** As senior postdoc, trained group members and led lab operations and instrument training.

Postdoctoral Associate Mar 2021 – Jul 2022
Sandia National Laboratories — Combustion Research Facility

- **Instrument Recovery:** Returned a relocated $1M+ cryogenic ion-trap mass spectrometer to full operation.
- **Lasers & Vacuum:** Restored the high vacuum system and 5K cold trap, calibrated a tunable IR OPO/OPA and UV dye lasers, and rebuilt timing and scan control in LabVIEW.

Graduate Research Assistant (PhD) May 2016 – Dec 2020
University of Missouri — Suits Group

- ***Science* Paper:** First author on quantum resonances in formaldehyde, measured by velocity-map imaging.
- **Laser Automation:** Designed laser automation and custom optical detection for UHV molecular-beam experiments; built a REMPI-characterized radical source.

Graduate Research Assistant Jun 2014 – Apr 2016
Wayne State University

- **Mass Spectrometry:** Developed ionization and ion-mobility mass spectrometry methods on Waters, Thermo, and Bruker platforms for biological and polymer analysis.
- **Lab Safety & SOPs:** Wrote the Synapt calibration SOP; laser-safety lead for the Class 3b UV laser.

EDUCATION

PhD, Physical Chemistry, University of Missouri Dec 2020
Dissertation: *Quantum Aspects of Roaming Dynamics* (Prof. Arthur G. Suits)

BS, Biochemistry, Saginaw Valley State University — Dow Chemical toxicology co-op (2010–2012) May 2013

Honors: Army Research Laboratory Journeyman Fellowship (2018) · WSU Thomas C. Rumble Fellowship (2014–2015) · WSU Chemistry Citation for Excellence in Teaching (2014) · SVSU President's Scholarship (2009–2013)

PUBLICATIONS & PRESENTATIONS

23 papers (8 first-author) and a 2026 preprint · 450+ citations · h-index 10 · cdfoley.com/publications

- **C. D. Foley** et al., "Site-specific photochemistry along a protonated peptide scaffold," *JACS* 146, 13282 (2024).
- **C. D. Foley** et al., "Molecular cage reports on its contents: spectroscopic signatures of cryo-cooled K^+- and Ba^{2+}-benzocryptand complexes," *J. Phys. Chem. A* 127, 6227 (2023).
- **C. D. Foley**, C. Xie, H. Guo, A. G. Suits, "Orbiting resonances in formaldehyde reveal coupling of roaming, radical, and molecular channels," *Science* 374, 1122 (2021).
- **C. D. Foley** et al., "Use of ion mobility spectrometry–mass spectrometry to elucidate architectural dispersity within star polymers," *ACS Macro Lett.* 4, 778 (2015).

Presentations (15+): Dynamics of Molecular Collisions Conference (2017, 2019, 2023), International Symposium on Molecular Spectroscopy (2022), ASMS Conference on Mass Spectrometry (2014, 2015), Army Research Laboratory Symposium (2018), ACS National Meeting (2013), and ARO and MURI program reviews.